

AYBERK YARKIN YILDIZ

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Summary

Researcher focusing on model-distributed inference via communication-aware pruning, mapping, and compression, and Markovian experimental design for robust re-training under concept drift. Experienced in building ML systems across sensing modalities, including wireless radar, RF signal sensing, sEMG fatigue analysis, and multivariate time-series modeling.

Education

Northeastern University <i>Ph.D. in Computer Engineering</i> <i>M.Sc. in Electrical and Computer Engineering</i>	<i>Boston, MA, USA</i> <i>Expected April 2028</i> <i>April 2025</i>
Bilkent University <i>B.Sc. in Electrical and Electronics Engineering</i>	<i>Ankara, Turkey</i> <i>July 2023</i>
Friedrich-Alexander-Universität <i>Erasmus Student in Elektrotechnik – Elektronik und Informationstechnik</i>	<i>Erlangen, Germany</i> <i>August 2022</i>
Mehmet Emin Resulzade <i>High School Degree</i>	<i>Ankara, Turkey</i> <i>June 2018</i>

Skills

Programming Languages: Python, MATLAB, C++
Frameworks & Tools: PyTorch, Tensorflow, PySpark, Slurm, Bash, Git, Linux, LaTeX
Languages: English (Fluent – C1 level), Turkish (Native Fluency), German (Beginner – A2 level)
Selected Courses: Advanced Machine Learning, Verifiable Machine Learning, Probabilistic System Modeling, Reinforcement Learning and Decision Making Under Uncertainty, Advanced Computer Vision, Machine Learning with Small Data

Experience

Northeastern University <i>Research Assistant</i>	<i>Boston, MA, USA</i> <i>Sep 2023 – April 2028</i>
Research Labs: DNAL, INSI, SPIRAL. Advisor: Prof. Stratis Ioannidis	
◦ Developed communication-aware distributed inference, Markovian experimental design, and 5G NR spectrum-sensing frameworks to address the bandwidth, energy, non-stationarity, and interference challenges of machine learning at the network edge, spanning optimization theory through deployment.	
OrthoKinetic Track <i>Data Science Intern</i>	<i>Boston, MA, USA</i> <i>Sep 2026 – Dec 2026</i>
◦ Developed machine learning pipelines for sEMG-based neuromuscular monitoring to enable objective assessment of performance, injury risk, and recovery across sports, rehabilitation, clinical, and defense applications.	
Neurocess Limited <i>Data Science / Machine Learning Engineer (Remote)</i>	<i>London, UK</i> <i>Sep 2022 – July 2023</i>
◦ Built few-shot Siamese CNN–Transformer architectures for multivariate sEMG time-series analysis to solve athlete performance monitoring under limited data, achieving similarity-based classification and anomaly detection across unseen motion patterns in PyTorch and Keras. View preprint 🔗	
KOCLAB, National Magnetic Resonance Research Center (UMRAM) <i>Undergraduate Researcher</i>	<i>Ankara, Turkey</i> <i>Sep 2021 – Sep 2022</i>
◦ Implemented deep learning models based on RNNs and Transformers to address missing data and temporal dependencies in biomedical time-series, producing imputation and forecasting pipelines validated in PyTorch experiments.	
TUBITAK SAGE <i>Intern</i>	<i>Ankara, Turkey</i> <i>June 2021 – July 2021</i>
◦ Designed nano-drone hardware and software simulations in Altium Designer to improve agility and reduce detectability in defense applications, supporting early-stage validation of stealth-oriented aerial platforms.	
UMRAM <i>Intern</i>	<i>Ankara, Turkey</i> <i>July 2020 – Aug 2020</i>
◦ Implemented and tested MATLAB-based interfaces for fundamental electronic devices (gaussmeter, analog filter, and DC power supply) to support scalable experimentation for future research, resulting in a publication in <i>Medical Physics</i> 🔗 .	

Projects

- Per-PRB Radar Detection and Localization in 5G NR Spectrograms** 2026 – Present
- Developed deep-learning-based spectrum-sensing models to localize radar emissions at per-resource-block resolution in 5G NR recordings, demonstrating reliable fine-grained detection under severe class imbalance and low-SNR conditions.
- Markovian Experimental Design under Concept Drift** 2024 – Present
- Implemented Kalman-filter-based data selection strategies to combat concept drift in sequential learning settings, demonstrating sustained predictive performance under non-stationary data distributions with provable optimality guarantees.
- Communication-Aware Distributed Inference with Compression and Pruning** 2024 – Present
- Developed communication-aware pruning, mapping, and latent-representation compression for distributed inference to cut latency and network overhead across heterogeneous edge devices, achieving up to $26\times$ speedup on Colosseum and Raspberry Pi testbeds with Lyapunov-based delay guarantees.
- Gradient Boosting Decision Trees on Medical Diagnosis** View publication [5]  2024
- Conducted large-scale empirical analysis of ensemble models to evaluate performance trade-offs against deep learning methods on tabular medical data, showing consistent superiority of GBDTs across multiple benchmarks.
- Wireless Radar Classification with Transformers** View publication [4]  2023 – 2024
- Implemented Transformer-based radar signal classifiers with LoRA and conformal prediction to improve robustness under distribution shifts, achieving reliable OOD performance in PyTorch.
- Portable RF Signal Sensing System Using SDR** View publication [6]  2023
- Implemented a GPU-accelerated SDR-based Electronic Support Measures (ESM) system to address real-time RF signal awareness requirements, enabling efficient detection, measurement, and classification using GNU Radio and XGBoost.
- Multivariate Time Series Imputation With Transformers** View publication [7]  2022
- Designed a Transformer-based autoencoder to solve missing-value imputation in multivariate time series, outperforming seven state-of-the-art methods by 13.5–50.5% over benchmark datasets.

Publications

- [1] **A. Y. Yıldız***, M. Sirera Perelló*, T. T. Davarakis*, J. Groen, H. Seferoglu, K. Chowdury, and S. Ioannidis, “CaMP: Communication-Aware Neural Mapping and Pruning.” (submitted to IEEE TMLCN)
- [2] **A. Y. Yıldız**, L. Su, C. Joe-Wong, E. Yeh, and S. Ioannidis, “Markovian Experimental Design under Concept Drift.” NeurIPS, 2026.
- [3] P. Gholami, T. T. Davarakis, T. Li, M. Sirera Perelló, S. Reddy, **A. Y. Yıldız**, A. Arora, A. Eryilmaz, S. Ioannidis, C. Li, H. Seferoglu, and N. Shroff, “Communication-Aware Model Distributed Inference via Latent Representation Compression.” ACM MobiHoc, 2026.
- [4] M. Belgiovine, J. Groen, M. Sirera, C. Tassie, **A. Y. Yıldız**, S. Ioannidis, and K. Chowdhury, “T-PRIME: Real-time Deployment of a Transformer-based Protocol Identification for Machine-learning at the Edge.” IEEE Transactions on Networking (2026).
- [5] **A. Y. Yıldız** and A. Kalayci, “Gradient boosting decision trees on medical diagnosis over tabular data.” 2025 IEEE International Conference on AI and Data Analytics (ICAD). IEEE, 2025.
- [6] G. S. Yavuz, B. Saygılı, Y. Aydınli, R. Dalkıran, İ. Eşin, M. Uluçay, B. Uykulu, S. S. Kıyma, O. Arikan, and **A. Y. Yıldız**, “Detection and classification architecture for sdr based radar electronic support measure systems.” 2024 32nd Signal Processing and Communications Applications Conference (SIU). IEEE, 2024.
- [7] **A. Y. Yıldız**, E. Koç, and A. Koç. “Multivariate time series imputation with transformers.” IEEE Signal Processing Letters 29 (2022): 2517-2521.

Honors & Awards

- Invited Talks: IEEE SPS, IEEE ICIP 2023 – 2024
- IEEE SPS’s top 25 downloaded articles 2022 – 2023
- Research Excellence Award at Bilkent University 2023
- High Honor Student & Tuition Scholarship at Bilkent University 2018 – 2023

Services

- Technical Reviewer: IEEE TMC, AAAI 2027, NeurIPS 2026, ICML 2026, AISTATS 2026, IEEE ICAD 2026
- Northeastern University SPIRAL Committee Member Dec 2024